UChicago REU Notes Week 2

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Contents

1 Week 2

1.1 Intro

For this portion, I want to start with some basic definitions covered by professor May in his office, about the definition of spectra / prespectra / generalized cohomology theory. I'll probably skip some of the details in the intro to make sure I can keep on track. (Also, I probably involve quite a lot of errors in between...I'll try to fix them)

1.1.1 Brown Representability Theorem, and Generalized Cohomology

Let $h\mathcal{C}$ denotes the homotopy category of based spaces (that admits a CW structure, as on them weak equivalences and homotopy equivalence are the same), then a functor $F : h\mathcal{C}^{\text{op}} \to \mathsf{Ab}$ is *representable*, if there exists a based space $Y \in h\mathcal{C}$, such that $F \cong [\_, Y]$.

Example 1.1. The *reduced cohomology functor* $\tilde{H}^n(\_; A)$ is one, for all $n \geq 0$ and a fixed abelian group A (as coefficient). Let $K(A, n)$ be the *Eilenberg-Maclane Space* of abelian group A and index $n \geq 0$ (i.e. the nth homotopy group $\pi_n(K(A, n)) = A$), then the following group isomorphism holds:

$$\tilde{H}^n(X; A) \cong [X, K(A, n)]$$

Using more adjunction property, since $\pi_n(A, n) = A = \pi_{n+1}(K(A, n+1)) = [S^{n+1}, K(A, n+1)] = [S^n \wedge S^1, K(A, n+1)] \cong [S^n, \Omega K(A, n+1)] = \pi_n(\Omega K(A, n+1))$, there is a homotopy equivalence $K(A, n) \xrightarrow{\simeq} \Omega K(A, n+1)$.

(Yeah, need to understand more about the details)

So, the regular cohomology we see on spaces, are one type of representable functors.

The classification of all such functors is given by *Brown Representability Theorem*:

Theorem 1.2. *Suppose the functor* $F : h\mathcal{C}^{\text{op}} \to \mathsf{Set}$ *satisfies:*

- $F(\bigvee_\alpha X_\alpha) \cong \prod_\alpha F(X_\alpha)$ *(i.e. maps coproduct to product).*
- *F maps* homotopy pushouts *(I believe this is the reduced mapping cylinder?) to* weak pullbacks *(omit the uniqueness of maps). This condition is equivalent to* Mayer-Vietoris axiom *(which seems like a weaker form of the gluing axiom in sheaf theory, and restrict to CW subcomplexes instead of open sets).*

Then, F is representable by some CW complex C, or $F \cong [\_, C]$.

The converse for the theorem also holds, as $[\_, C]$ also satisfies the above axioms (need more details here).

Here is a definition of generalized cohomology:

Definition 1.3. A *generalized cohomology*, is a sequence of spaces $\{E_n\}_{n\geq 0}$, with homotopy equivalences $\{\epsilon_n : E_n \xrightarrow{\simeq} \Omega E_{n+1}\}_{n\geq 0}$.

In a more explicit form, $E^n(X) := [X, E_n]$ has all $n \geq 0$ satisfies $E^n(X) = [X, E_n] \cong [X, \Omega E_{n+1}]$, which is naturally a group (and an abelian group for $n \geq 2$, due to the abelian group structure on $[X, \Omega^m Y]$ for $m \geq 2$).

I remembered the motivation is that, by Brown Representability Theorem, if a sequence of representable functors $E^* : h\mathcal{C}^{\text{op}} \to \mathsf{Set}$ satisfy similar properties to ordinary cohomology (for instance, the connecting morphism $E^n \to E^{n+1}$, and excision axiom?), then Brown Representability Theorem guarantees it must be the above form. So, they're precisely all the generalized cohomology.

1.1.2 Ω-Prespectra, Prespectra, and Spectra

When talking about generalized cohomology, we have a sequence of based spaces $\{E_n\}_{n\geq 0}$, together with homotopy equivalences $\epsilon_n : E_n \xrightarrow{\simeq} \Omega E_{n+1}$, which is a prototype of what's called Ω-*Prespectra*.

In general, fixing a based space $X \in h\mathcal{C}$, the sequence $\{\Sigma^n X\}_{n\geq 0}$ has a collection of natural maps $\{\text{id}_n : \Sigma(\Sigma^n X) \to \Sigma^{n+1} X\}_{n\geq 0}$, which the adjunction (Σ, Ω) guarantees a collection of maps $\{\eta_n : \Sigma^n X \to \Omega(\Sigma^{n+1} X)\}_{n\geq 0}$. In general these maps are not homotopy equivalence, so these type of sequences are a type of *Prespectra*, called *suspension prespectra.* More generally, *Prespectra* is a sequence of spaces $\{X_n\}_{n\geq 0}$, with morphisms $\{\mu_n : X_n \to \Omega X_{n+1}\}_{n\geq 0}$.

For the strongest form possible, if such map $\epsilon_n : E_n \to \Omega E_{n+1}$ is a homeomorphism, then these are called *Spectra.*

Professor May restricts to three specific categories:

- Let $\mathcal{P}$ denotes the *category of prespectra.*
- Let $\mathcal{Q}$ denotes the *category of inclusion prespectra*, meaningeach defining morphism $\mu_n : X_n \to \Omega X_{n+1}$ is an inclusion.
- Let $\mathcal{S}$ denotes the *category of spectra.*

For each category, we require the morphism $\phi : X. \to T.$ to be a sequence of based maps $\phi_n : X_n \to T_n$, such that the following diagram commutes:

$$\begin{tikzcd}
\Sigma X_n \arrow[r, "\Sigma\phi_n"] \arrow[d, "\mu'_{X_n}"'] & \Sigma T_n \arrow[d, "\mu'_{T_n}"] \\
X_{n+1} \arrow[r, "\phi_{n+1}"'] & T_{n+1}
\end{tikzcd}$$

where the map $\Sigma X_n \to X_{n+1}$ is based on the adjoint property $[\Sigma X_n, X_{n+1}] \cong [X_n, \Omega X_{n+1}]$. Here professor May had talked about how the prototype of the morphism only requires it to commute up to homotopy, but turns out it's too weak as a condition.

With these morphisms, there is an obvious forgetful functor $\mathcal{S} \to \mathcal{Q} \to \mathcal{P}$. It seems to satisfy all the conditions for Freyd's Adjoint Functor Theorem to hold, both of the forgetful functors have a left adjoint.

Here, the left adjoint of $\mathcal{Q} \to \mathcal{P}$ is hard to construct, yet there is a clear construction for the left adjoint of $\mathcal{S} \to \mathcal{Q}$, denoted as $L : \mathcal{Q} \to \mathcal{S}$, by $T. \mapsto (LT).$, where $(LT)_n := \operatorname{colim}_q \Omega^{n+q} T_q$ (due to the inclusion $T_q \to \Omega T_{q+1}$, it induces $\Omega^{n+q} T_q \to \Omega^{n+(q+1)} T_{q+1}$, hence having a sequence of direct system).

Professor May had claimed the homeoomorphism $(LT)_n \cong \Omega(LT)_{n+1}$ due to the fact that Ω commutes with colimit (I do need to work out the detail, as normally Ω as a right adjoint of Σ would be thought of preserving limits instead of colimits).

Most part of the following notesare from professor May's *Infinite Loop Space Theory* paper (https://www.math.uchicago.edu/ may/PAPERS/18.pdf).

1.2 Spectra, Infinite Loop Spaces, and Stabilization Functor

If require spectra to be a sequence of spaces $E = \{E_n\}_{n\geq 0}$ with homeomorphism $E_n \cong \Omega E_{n+1}$, we have the following definition:

Definition 1.4. A space X that has a homotopy type of E_0 for some spectra E is called an *infinite loop space.*

If X, Y are infinite loop spaces of spectra E, F respectively, then a map $f : X \to Y$ is caled an *infinite loop map*, if it's induced by some morphism of spectra $\tilde{f} : E \to F$.

Also, we have the notion of stable homotopy groups:

Definition 1.5. Given a space X, its*nth stable homotopy group* $\pi_n^s(X) := \mathrm{colim}_j \pi_{n+j}(\Sigma^j X)$.

Here, the colimit is due to the natural map $\Sigma(\Sigma^j X) \to \Sigma^{j+1} X$, inducing $\Sigma^j X \to \Omega(\Sigma^{j+1} X)$. Hence, the following isomorphism holds:

$$\pi_k(\Sigma^j X) \to \pi_k(\Omega(\Sigma^{j+1} X)) = [\Sigma^k S^0, \Omega(\Sigma^{j+1} X)] \cong [\Sigma^{k+1} S^0, \Sigma^{j+1} X] = \pi_{k+1}(\Sigma^{j+1} X)$$

Hence, fixing $k := n + j$, this induces a map $\pi_{n+j}(\Sigma^j X) \to \pi_{n+(j+1)}(\Sigma^{j+1} X)$.

I remembered part of the reason this is called stable homotopy group, is because for large enough j this direct system in fact stabilizes.

For stabilization, it's a functor $QX := \mathrm{colim}_q \Omega^q \Sigma^q X$. Which, it has the following isomorphism on homotopy groups:

$$\pi_n(QX) \cong \pi_n^s(X)$$